

FINAL PROJECT REPORT [WEEK 4]

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Group Number: Group 3

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Calendar Week: CW43

Date: [Oct 27 – Nov 2, 2025]

Progress Summary

This week, the Auto Robot was completed and now in the code testing and improvement phase. So, this week we will focus on the Manual Robot, specifically the launcher mechanism, which is really challenging. The first design used a TT motor to launch the ball and successfully launched the 70mm diameter ball about 450mm, but was unable to launch the 100mm diameter ball.

After multiple adjustments to the arm length and angle of freedom. We are still unable to launch a big ball. Therefore, we decided to focus on launching small balls since there are more small balls than larger ones. The current Manual Robot Version 2 is completed, which still requires optimization on the throwing arm.

Next, our team plans to replace the aluminum launcher arm with a fully 3D printed arm instead for lighter weight and to improve the performance. Both robots are expected to be finalized and tested on the field next week.

Next Week Plan

- Complete both Auto and Manual Robots and test them on the field.
- Replace aluminum launcher arm with 3D printed version.
- Optimize throwing performance and adjust strategy based on testing results.

Problems Encountered

- The TT motor generated not enough torque for the launcher to launch the larger balls.
- Arm adjustment and angle tuning did not improve the range.
- Strategy had to be adapted if the robot was unable to launch a larger ball.